DO NOW — RECALL

Lesson 107 · Paging, Segmentation and Virtual Memory

Name: _____ Class: _____ Date: _____

1. Identify two services provided by an OS.

2. What is the time complexity of sequential search in the worst case?

3. After pushing A then B onto a stack, which is popped first?

4. Why must a binary search operate on ordered data?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Any two: process management, memory management, files, devices, security or interface.

2. Model answer

O(n).

3. Model answer

B, because a stack is LIFO.

4. Model answer

It discards half the search space using an ordering comparison.

By the end, you can…

• Virtual memory is split into fixed-size pages; RAM into frames.

• Logical variable-size segments represent code, data or stack regions.

• The OS maps addresses and can move inactive pages to secondary storage.

Vocabulary — know these terms

page — Fixed-size virtual-memory block.

frame — Fixed-size physical-RAM allocation unit.

segmentation — Memory organised into variable-size logical segments.

page fault — Access to a page not currently resident in RAM.

NEW LEARNING

Read, annotate and explain

Paging

Virtual memory is split into fixed-size pages; RAM into frames.

Segmentation

Logical variable-size segments represent code, data or stack regions.

Virtual memory

The OS maps addresses and can move inactive pages to secondary storage.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

A 2,500-byte virtual address uses 1,024-byte pages. Find the page and offset, then place it in frame 7.

Model answer

Page number = 2 and offset = 452. Frame 7 base is 7 × 1,024 = 7,168; physical address = 7,620, assuming that page is mapped to frame 7.

Check: link each claim to the scenario and use precise OCR terminology.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Differentiate paging and segmentation, with one reason for using each. [6]

Question 2

With 4,096-byte pages, calculate page number and offset of virtual address 8,193. [4]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Explain why heavy paging can cause thrashing and slow down a computer. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Virtual memory increases the speed of every memory access.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.